A KOSZUL-REGULAR IMMERSION OF SPACES IS A LOCAL COMPLETE INTERSECTION MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 11271-11275 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 11277-11287, the complete author proof (9 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Counted B side; schemes twin is the cited More on Morphisms lemma itself (outside pool).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. In-source definitions regular-immersion and lci; More on Morphisms, lemma regular-immersion-lci.

Source reference. Original lemma 069F: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 11153–11164. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 069L: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 11287–11287. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. A Koszul-regular immersion of algebraic spaces over S is a local complete intersection morphism.*

Proof. Let $i : X \to Y$ be a Koszul-regular immersion of algebraic spaces over S. By definition there exists a surjective étale morphism $V \to Y$ where V is a scheme such that $X \times_Y V$ is a scheme and the base change $X \times_Y V \to V$ is a Koszul-regular immersion of schemes. By More on Morphisms, Lemma 069L we see that $X \times_Y V \to V$ is a local complete intersection morphism. From Definition 069F we conclude that i is a local complete intersection morphism of algebraic spaces. □

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.